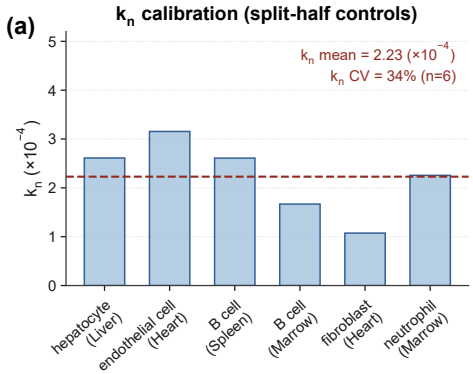
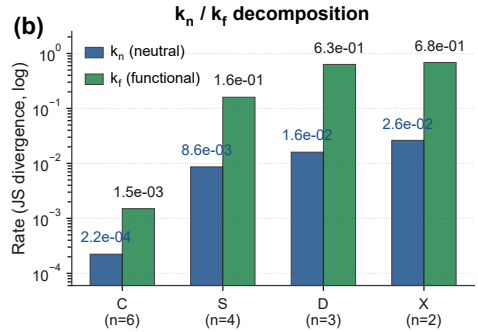
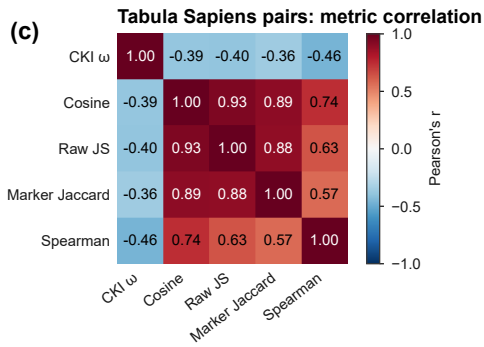
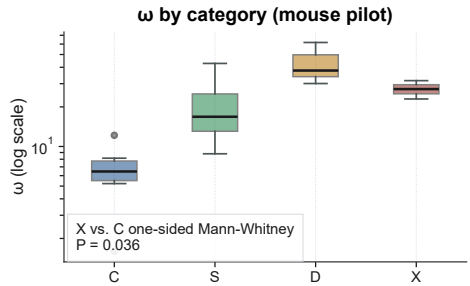
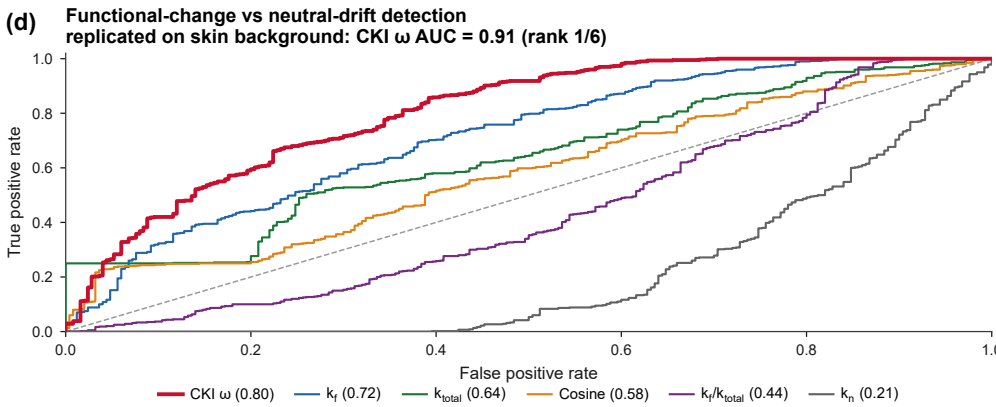
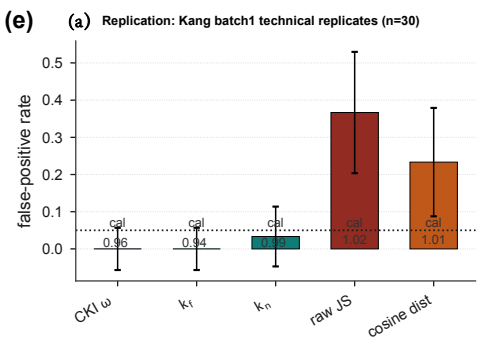

(a)
k_n calibration (split-half controls)
k_n mean = 2.23 (×10^{-4})
k_n CV = 34% (n=6)
k_n (×10^{-4})
hepatocyte (Liver)
endothelial cell (Heart)
B cell (Spleen)
B cell (Marrow)
fibroblast (Heart)
neutrophil (Marrow)

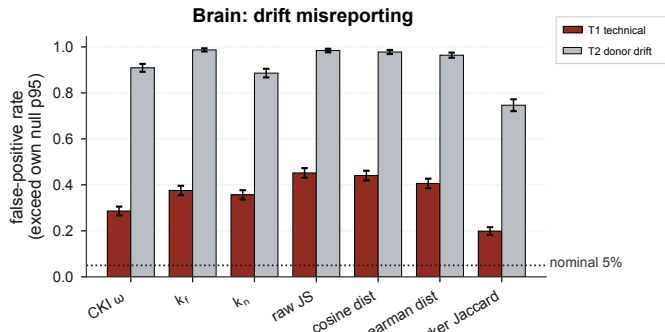

(b)
k_n / k_f decomposition
k_n (neutral)
k_f (functional)
Rate (JS divergence, log)
2.2e-04
1.5e-03
8.6e-03
1.6e-01
1.6e-02
6.3e-01
2.6e-02
6.8e-01
C (n=6)
S (n=4)
D (n=3)
X (n=2)

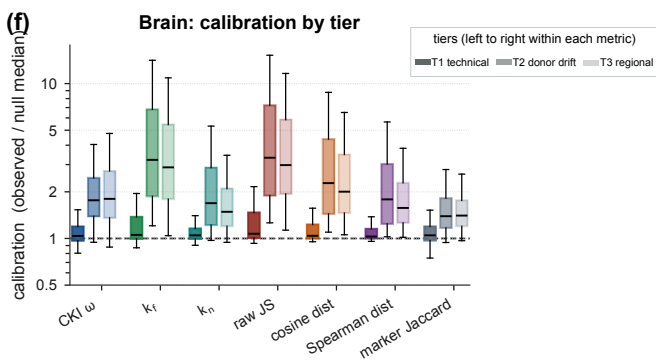

(c)
Tabula Sapiens pairs: metric correlation
CKI ω
Cosine
Raw JS
Marker Jaccard
Spearman
1.00 -0.39 -0.40 -0.36 -0.46
-0.39 1.00 0.93 0.89 0.74
-0.40 0.93 1.00 0.88 0.63
-0.36 0.89 0.88 1.00 0.57
-0.46 0.74 0.63 0.57 1.00
Pearson's r


ω by category (mouse pilot)
ω (log scale)
X vs. C one-sided Mann-Whitney
P = 0.036
C
S
D
X


(d)
Functional-change vs neutral-drift detection
replicated on skin background: CKI ω AUC = 0.91 (rank 1/6)
True positive rate
False positive rate
CKI ω (0.80)
k_f (0.72)
k_{total} (0.64)
Cosine (0.58)
k_f/k_{total} (0.44)
k_n (0.21)


(e)
(a) Replication: Kang batch1 technical replicates (n=30)
false-positive rate
cal 0.96
cal 0.94
cal 0.06
cal 1.02
cal 1.01
CKI ω
k_f
k_n
raw JS
cosine dist


Brain: drift misreporting
T1 technical
T2 donor drift
false-positive rate (exceed own null p95)
nominal 5%
CKI ω
k_f
k_n
raw JS
cosine dist
Spearman dist
marker Jaccard


(f)
Brain: calibration by tier
tiers (left to right within each metric)
T1 technical
T2 donor drift
T3 regional
calibration (observed / null median)
CKI ω
k_f
k_n
raw JS
cosine dist
Spearman dist
marker Jaccard